

Functions

5.1 Main Function

Functions are vital in programming development. For the sake of convenience and simplicity, a program is divided into smaller units called functions. This is one of the major principles of sequential structural programming. However, the advantage of using functions in a program is to reduce the size of the program. The functions are called at different phases of the program to simplify its execution.

The `main()` function generally returns a value of type `int`. The language defines the `main()` method and matches one of the following prototypes. These are as follows:

```
int main();  
int main(int argc, char * argv[]);
```

It is to be noted that functions returning a value should always use the return statement. Hence, whenever we declare and define a function, it should be declared as follows:

```
int main()  
{  
..... statements .....  
.....statements.....  
  
return 0;  
  
}
```

By default, the return functions is of type `int` and is considered optional in the `main()` header of a program. For any function that has a return value, there should be a `return` statement with-